

ViTacPhys: Physical Property-Aware Grasping from Human Visual-Tactile Demonstrations

Anonymous

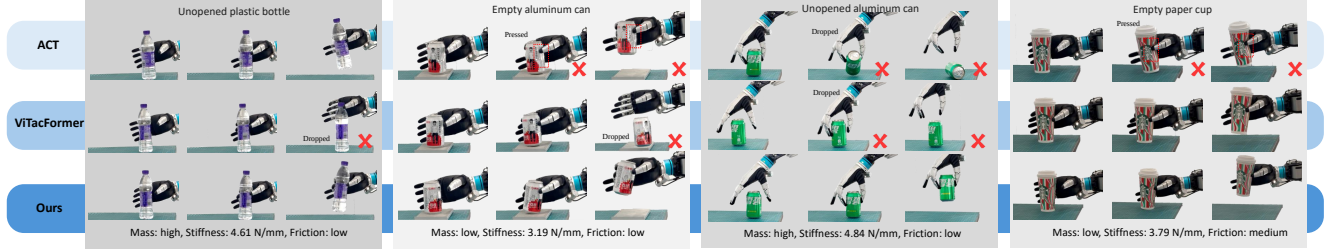


Fig. 1. Compared with ACT and ViTacFormer, ViTacPhys conditions the robot policy on predicted mass, stiffness, and friction, enabling stable grasps across visually similar objects with different manipulation requirements.

Abstract—Object physical properties such as mass, stiffness, and friction directly determine how a robot should grasp, support, and manipulate an object, yet they are often hidden from static visual observations and difficult to estimate before contact. Humans naturally combine visual priors with dense tactile feedback during interaction to infer these properties and adapt their actions. Motivated by this capability, we introduce H2Touch, a human visual-tactile manipulation dataset collected with a wearable sensing system that synchronizes wrist-mounted RGB observations and fingertip piezoresistive tactile maps. H2Touch covers 60 everyday objects from 7 categories and provides explicit annotations for mass, structural stiffness, and static friction coefficient. Based on this dataset, we propose a temporal multimodal physical-property predictor that fuses RGB frames, tactile pressure maps, visual optical flow, tactile flow, and language-derived semantic priors to classify mass and friction ranges and regress structural stiffness during grasping interactions. We further study human-to-robot transfer by adapting the predictor to a dexterous robot and using its outputs as structured property tokens for an ACT-style adaptive grasping policy. Experiments show that the proposed predictor generalizes to unseen objects under multiple split settings, transfers to robot observations with limited robot data, and improves real-robot grasp success and force adaptation, especially for visually similar out-of-distribution objects with different physical properties. H2Touch provides a practical bridge between human multimodal interaction, physical-property perception, and property-aware robot policy learning.

Index Terms—Vision-tactile fusion, physical property prediction, adaptive grasping

I. INTRODUCTION

Recent Vision-Language-Action (VLA) models [?] and World Action Models (WAMs) [?] have shown impressive capability in dexterous and complex manipulation tasks. However, robots operating in real environments, such as logistics robots handling packages or household robots tidying tables, must do more than recognize object category, shape, and pose. They also need to understand physical properties that directly determine the control strategy during interaction. For example, mass affects the supporting force required during grasping and transport [?], [?], [?]; friction determines whether an object is likely to slip [?], [?], [?], [?];

and stiffness influences the grasping contact force as well as the gripper aperture or dexterous-hand joint configuration [?], [?], [?]. When manipulating a delicate paper cup, excessive force may crush or deform the object, whereas insufficient force may cause it to slip and fall [?]. Humans have rich prior knowledge about the physical properties of common objects and the corresponding manipulation strategies [?]. Even for unseen objects, humans can usually combine visual experience with dense tactile feedback, often from a brief touch or a light hefting motion, to quickly infer the object’s physical properties and adjust the manipulation action accordingly [?], [?], [?]. For robots, however, acquiring such cross-modal priors or rapidly estimating physical properties that cannot be directly observed remains a central challenge for general object manipulation [?], [?].

Prior work estimates physical properties either from priors, such as RGB or tactile representation learning [?], [?], [?], [?], or from designed interactions such as pushing, pulling, sliding, poking, and controlled physical experiments [?], [?], [?], [?], [?], [?], [?], [?]. These settings provide useful evidence, but they are often tied to specific robot actions or remain offline estimators. Human manipulation data is a promising source of rich physical cues [?], [?], [?], yet it is rarely transferred to real robot execution or converted into policy inputs for adaptive manipulation [?], [?].

We therefore introduce H2Touch, a human visual-tactile manipulation dataset with explicit labels for mass, structural stiffness, and static friction coefficient. We learn these properties from temporal wrist RGB observations, fingertip piezoresistive tactile maps, visual/tactile flow, and language-derived semantic priors. To bridge human demonstrations and robot execution, we combine RGB-domain replacement with co-training on limited robot data. The predicted properties are then used as structured tokens for an ACT-style grasping policy, enabling force and action adaptation to hidden physical differences.

The main contributions of this paper are summarized as follows:

- H2Touch, a human visual-tactile dataset with explicit

mass, stiffness, and friction labels.

- A temporal visual-tactile-language predictor that transfers from human demonstrations to real robot observations.
- A property-conditioned imitation policy that uses predicted physical properties for adaptive dexterous grasping.

II. RELATED WORK

Object Physical Properties Datasets. Physical-property datasets can be broadly divided into two categories: datasets with explicit labels and datasets with implicit labels. Representative datasets are summarized in Table ?? . Implicit datasets are often larger in scale, but they are usually tied to specific experimental scenarios. For example, Physics 101 collects observations from controlled setups such as ramp experiments [?]. Explicit datasets require specialized protocols and instruments to obtain ground-truth physical properties; therefore, they inevitably contain measurement noise and are typically smaller. As shown in Table ?? , existing datasets cover diverse modalities, and in some cases the modalities can be combined to obtain a theoretical solution. For instance, position information from depth or motion-capture modalities, together with tactile force, can be used to infer elastic properties. In addition, most existing real-robot experiments use parallel grippers, and high-precision vision-based tactile sensors such as GelSight are difficult to mount on dexterous hands. Their tactile modalities are also commonly represented as direct force inputs [?], [?], [?], [?], while only a few works use tactile images as model inputs [?], [?]. Table ?? also shows that mass-related properties, stiffness-related properties, and friction-related properties are among the most manipulation-relevant attributes. Our work predicts mass, structural stiffness, and static friction coefficient through dexterous-hand manipulation.

Learning Object Physical Properties. Physical properties can be estimated from visual priors, tactile evidence, or interactive dynamics. Vision-based methods infer mass from RGB, object geometry, or density priors [?], [?], but they are sensitive to hidden fillings and unfamiliar materials. Tactile and force-based methods provide contact evidence for mass, stiffness, hardness, and friction through designed hefting, sliding, pushing, or squeezing motions [?], [?], [?], [?], [?], [?], [?], [?], [?], [?]. Multimodal methods improve robustness by combining visual priors with contact feedback, and recent work also uses LLM/VLM priors for physical reasoning [?], [?], [?].

Self-supervised approaches learn physical representations by aligning RGB with tactile images or force signals [?], [?], [?], or by predicting interaction outcomes such as optical flow, pushing trajectories, swing angles, or post-contact images [?], [?], [?], [?], [?], [?]. These methods reduce annotation cost but often learn implicit properties tied to a specific action or setup. Human demonstrations contain natural adaptations to hidden physical properties [?], [?], [?], but prior work rarely transfers these cues to real robots or

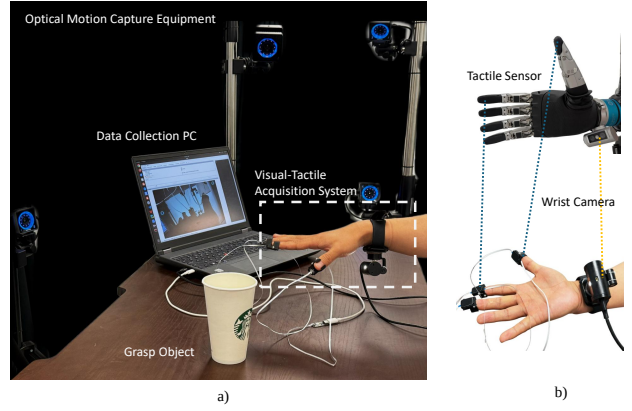


Fig. 2. Human visual-tactile data collection setup. The system synchronizes wrist RGB video, fingertip tactile maps, and motion-capture measurements while participants grasp everyday objects.



Fig. 3. Objects in H2Touch. The dataset covers diverse containers, toys, fruits, sponges, and soft objects with varied mass, stiffness, and surface friction.

exposes them as structured priors for policy learning. Our work targets this missing link.

Adaptive Manipulation by Imitation Learning. Adaptive imitation learning that integrates tactile or force information makes fine-grained manipulation more feasible. Existing multimodal fusion mechanisms include self-attention [?] and cross-attention [?], while tactile images can be incorporated through MoE architectures [?] or contrastive learning with other modalities [?]. Within the ACT framework [?], FTACT [?] simply adds raw force signals and shows that force differences across manipulation stages can sensitively indicate different execution strategies. ViTacFormer [?] additionally learns to predict the next tactile frame. SoftGrasp [?] is closely related to our application: it proposes adaptive grasping for objects with different softness and hardness and demonstrates generalization over a range of objects. In contrast, our work explicitly predicts manipulation-relevant physical properties from human visual-tactile demonstrations and uses them as structured priors for downstream adaptive grasping.

III. MULTIMODAL PHYSICAL PROPERTY DATASET

We propose H2Touch, a large-scale in-house dataset for physical-property prediction from human visual-tactile manipulation. This section introduces the synchronized data-collection and annotation pipeline and then summarizes the dataset statistics.

TABLE I
PHYSICAL-PROPERTY DATASETS.

Implicit Labels		
Dataset	#Obj.	Properties / Modalities
Physics 101 [?]	101	Mass, density, elasticity; RGB-D video
OmniPush [?]	250	Mass distribution, friction; RGB-D, motion, state
Synesthesia [?]	400+	Material, friction, compliance; RGB, tactile signals
Fabrics [?]	118	Material, stiffness, thickness; RGB-D, GelSight

Explicit Labels		
Dataset	#Obj.	Properties / Modalities
Image2Mass [?]	150K/56	Volume, density, mass; RGB, bounding box
Tactile glove [?]	26	Object shape, mass; whole-hand tactile glove
Octopi [?], [?]	35	Hardness, elasticity, roughness; RGB, GelSight, language
H2Touch (Ours)	60	Mass, stiffness, friction; wrist RGB, tactile maps

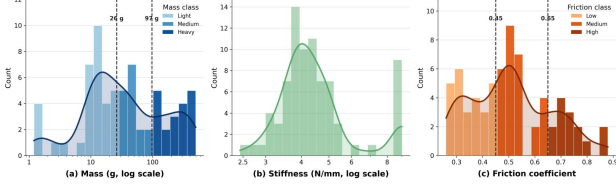


Fig. 4. Physical-property label distributions. Mass and friction are discretized into three ranges for classification, while structural stiffness is retained as a continuous regression target.

A. Data Collection

The dataset is collected with a wearable visual-tactile acquisition system designed for human demonstrations as shown in Figure ???. The system consists of a tactile sensing glove and a wrist-mounted RGB camera. The tactile subsystem is a customized wearable glove instrumented with three fingertip tactile sensor arrays mounted on the thumb, index finger, and middle finger. Visual observations are recorded by the wrist-mounted camera. We use the same type of visual sensor as the real robot and tactile sensors with the same spatial density distribution as those mounted on the robot hand. In addition, motion-capture markers are attached near the corresponding fingertips so that fingertip displacement can be measured inside the motion-capture workspace.

Each fingertip tactile array produces a pressure image according to the tactile taxel layout, and tactile readings from the three fingers are concatenated and post-processed into $I_{\text{tac}} \in \mathbb{R}^{180 \times 40}$. The wrist-mounted camera stream is post-processed into RGB images $I_{\text{visual}} \in \mathbb{R}^{160 \times 90 \times 3}$. After calibrating the raw sensor readings, we can also obtain the contact force F from the tactile measurements. Compared with a head-mounted viewpoint, the wrist-mounted camera provides a finer grasp-centric perspective and more clearly observes the hand, the object appearance, object pose, and contact-induced deformation during grasping. To increase data diversity, we only require collectors to align the glove and wrist camera at the visual scale during setup. Visual and tactile streams are aligned by timestamps at 30 Hz.

To compensate for the limitation that piezoresistive tactile sensors do not directly measure tangential force, we collect data using two interaction protocols. The first is a vertical grasping protocol: the object is placed on a table, pinched by the fingertips, and then lifted vertically at an approximately constant speed. This protocol yields relatively stable tactile signals after contact, and object physical properties are reflected throughout the grasping process. The second is a shaking-grasp protocol: after grasping the object, the

participant swings it laterally with a large amplitude. Under this protocol, tactile signals vary during rotation. When the object rotates such that the normal direction of the fingertip piezoresistive taxels becomes parallel to gravity, the tactile readings provide useful evidence for estimating mass. For each object, we collect ten trials and include multiple grasp instances with different approach directions and hand postures to increase interaction diversity. More details are provided in the Appendix.

B. Data Annotation

Our method predicts three physical properties of the grasped object: mass, structural stiffness, and surface static friction coefficient. We build an annotation pipeline to label these properties. Object mass is measured using a precision laboratory scale with a resolution of 0.1 g.

Structural stiffness is measured inside the motion-capture workspace while the participant wears the tactile glove with fingertip markers. Fingertip normal displacement is obtained from the motion-capture markers, while fingertip normal contact force is obtained from the calibrated tactile sensor readings after applying a fourth-order Butterworth filter. Given multiple displacement-force pairs $\mathbf{x}$ and $\mathbf{F}$, we fit a linear force-displacement model and compute the least-squares stiffness estimate during pinching as $k^{(\text{ls})} = \text{cov}(\mathbf{x}, \mathbf{F}) / \text{var}(\mathbf{x}) = \sum_i (x_i - \bar{x})(F_i - \bar{F}) / \sum_i (x_i - \bar{x})^2$. The unit of structural stiffness is N/mm.

The static friction coefficient is measured by placing the object on a silicone-coated plane whose surface finish matches that of the glove sensing layer. We gradually increase the inclination angle until the object reaches the critical point of sliding. The critical angle θ is recorded, and the static friction coefficient is computed as $\mu_s = \tan(\theta)$. For each object, each measurement protocol is repeated five times, and the mean value is used as the final label. We also compute the sample standard deviation to represent label uncertainty. Additional annotation details are provided in the Appendix.

C. Dataset Overview

Using the above pipeline, we construct the H2Touch dataset, which contains 60 everyday objects shown in Figure ??. Each object is annotated with all three physical properties described above. Although mass, stiffness, and friction coefficient are collected as continuous values, our hardware does not provide tangential tactile feedback, so the model input is less sufficient for estimating mass and static

friction coefficient precisely. In addition, for downstream tasks such as grasping, these two properties mainly guide policy behavior through property ranges rather than exact numerical values. To make downstream prediction more stable, we cluster the raw mass and static-friction values of all objects into three categories and post-process the labels as low, medium, and high. Structural stiffness is kept as an absolute continuous value. The distributions of the three physical properties in the dataset are shown in Figure ???. The raw data distributions and detailed split thresholds are provided in the Appendix.

IV. METHOD

Our method learns object physical properties from human visual-tactile grasping interactions and uses the predicted properties as structured priors for adaptive robot grasping. The pipeline consists of a temporal visual-tactile property predictor, a VLM-distilled semantic prior, and a physical-property-conditioned ACT policy shown in Figure ??.

A. Problem Formulation

For each interaction trial, we observe a post-processed wrist-mounted RGB sequence $I_{\text{visual}} \in \mathbb{R}^{T \times 160 \times 90 \times 3}$ and a post-processed fingertip tactile sequence $I_{\text{tac}} \in \mathbb{R}^{T \times 180 \times 40}$. The predictor M_{phys} outputs mass logits $m \in \mathbb{R}^3$, friction logits $f \in \mathbb{R}^3$, and a continuous stiffness prediction $s \in \mathbb{R}$. As described in Sec. ??, mass and friction are discretized into low, medium, and high ranges, while stiffness remains continuous.

We sample a temporal window $W = \{t, t+1, \dots, t+L-1\}$ with $L = 15$. Adjacent post-processed frames are used to compute visual flow $F_v^\tau = \Phi_{\text{flow}}(I_v^\tau, I_v^{\tau+1})$ and tactile flow $F_t^\tau = \text{Rep}_3(I_t^{\tau+1} - I_t^\tau)$. These flow inputs summarize hand-object motion and tactile pressure changes during grasping, lifting, and shaking.

B. Visual-Tactile Encoders

For both visual and tactile modalities, we use a two-stream encoder that separates static content from temporal motion. The content stream extracts per-frame features from the resized RGB frames or tactile pressure maps, and the motion stream extracts features from the corresponding visual or tactile flow. Both streams use ResNet-18 backbones followed by a projection layer to produce 256-dimensional features, i.e., $H_v^c = E_v^c(I_v)$, $H_v^m = E_v^m(F_v)$, $H_t^c = E_t^c(I_t)$, and $H_t^m = E_t^m(F_t)$, where $H^c, H^m \in \mathbb{R}^{B \times L \times 256}$ after temporal alignment. For each modality, content and motion features are concatenated and passed through a GRU to aggregate temporal interaction information, giving $z_v = \text{GRU}_v([H_v^c; H_v^m])$ and $z_t = \text{GRU}_t([H_t^c; H_t^m])$, where $z_v, z_t \in \mathbb{R}^{B \times 512}$. This design lets the model use object appearance and tactile pressure distributions together with their temporal changes, which are essential for inferring mass, stiffness, and friction during manipulation.

C. VLM-Distilled Semantic Prior

Visual and tactile observations provide direct evidence during interaction, but generalization to unseen objects also benefits from semantic priors about likely material and physical properties. We therefore distill a textual prior from a vision-language model (VLM). For each temporal window, we uniformly sample five RGB frames, $\{I_k^{(j)}\}_{j=1}^5 = \text{UniformSample}(\{I_{k,\tau}\}_{\tau=t}^{t+L-1}, 5)$. The VLM is prompted to describe the manipulated object, its visible interaction state, and plausible mass, stiffness, and friction ranges with confidence-aware alternatives. To reduce overconfident visual-only guesses, the prompt includes both positive and negative in-context examples, with full prompt details provided in the Appendix.

The generated text is encoded at two levels. A Sentence Transformer produces a global sentence feature $e_{\text{sent}} \in \mathbb{R}^{768}$, and a BERT encoder produces word-level features $E_{\text{word}} \in \mathbb{R}^{N \times 768}$. We concatenate the sentence token with word tokens and project them to the fusion dimension as $F_{\text{text}} = \text{LN}(W_{\text{text}}[e_{\text{sent}}; E_{\text{word}}]) \in \mathbb{R}^{B \times (N+1) \times 256}$. From a probabilistic perspective, this text feature acts as a semantic prior that constrains the physical-property posterior:

$$p(y | X_v, X_t, F_{\text{text}}) = \frac{p(X_v, X_t | y, F_{\text{text}})p(y | F_{\text{text}})}{p(X_v, X_t | F_{\text{text}})} \quad (1)$$

where y denotes one of the target physical properties and X_v, X_t denote visual and tactile observations.

D. Multimodal Fusion and Prediction Heads

The fusion module combines three sources of information: visual interaction features, tactile interaction features, and VLM-derived textual priors. We first apply bidirectional cross-attention between the visual feature z_v and tactile feature z_t so that each modality can select complementary cues from the other. The attended visual and tactile features are concatenated and projected into a compact visual-tactile representation z_{vt} .

We then inject the semantic prior with a gated cross-attention module. The text feature attends to the visual-tactile feature to produce z_{text} , and a learned gate controls how much semantic prior should be added:

$$\begin{aligned} z_{\text{text}} &= \text{CA}(z_{\text{vt}}, F_{\text{text}}, F_{\text{text}}), \\ z_{\text{fused}} &= [z_{\text{vt}}; \sigma(W_g[z_{\text{vt}}; z_{\text{text}}]) \odot z_{\text{text}}]. \end{aligned} \quad (2)$$

The gate prevents unreliable text priors from overwhelming direct sensor evidence. The final feature z_{fused} is passed to three lightweight prediction heads: a mass head and a friction head output ordered three-class logits, while a stiffness head outputs a scalar stiffness value. This keeps the three physical-property predictions coupled through the shared multimodal representation while allowing each property to use a task-specific output layer.

E. Training Loss

Mass and static friction are continuous quantities discretized into ordered ranges, so predicting an adjacent range

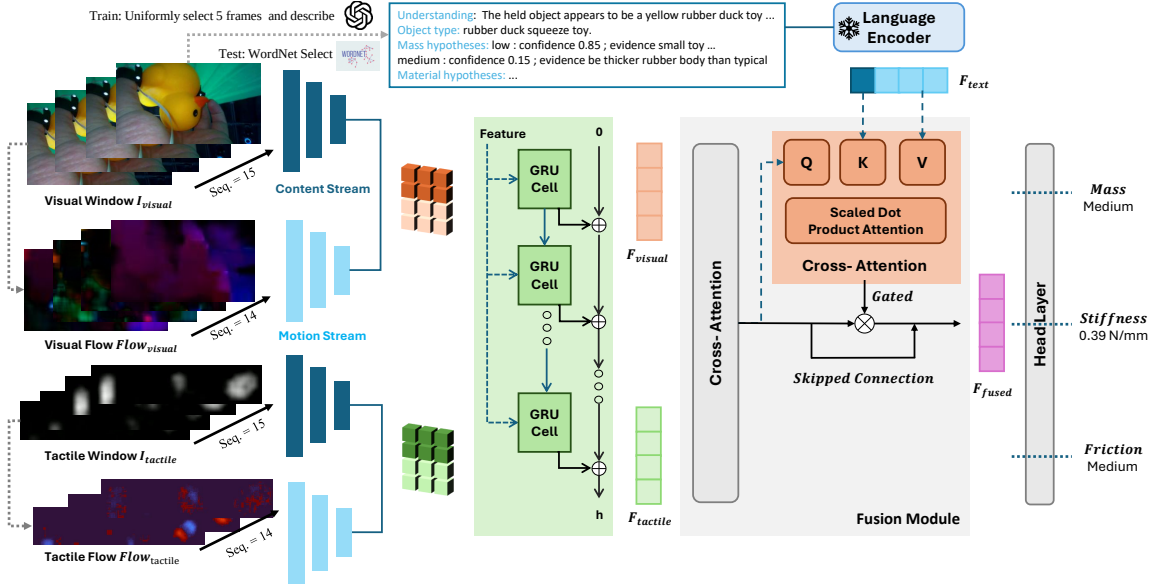


Fig. 5. Visual-tactile physical-property predictor. RGB frames, tactile maps, visual flow, tactile flow, and language-derived semantic priors are fused by temporal encoders and gated cross-attention to predict mass, stiffness, and friction.

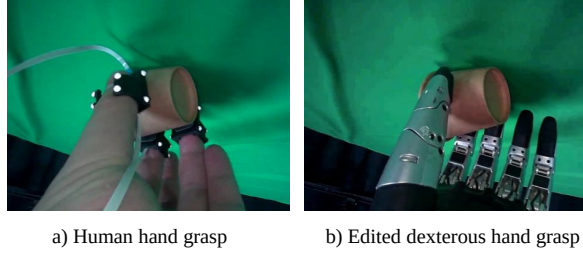


Fig. 6. RGB-domain replacement for human-to-robot transfer. Human-hand grasp videos are translated into robot-hand observations while preserving object identity and interaction state.

should be penalized less severely than predicting a distant range. We therefore use ordinal regression: each label y is converted into threshold labels $r_k = 1[y > k]$, and the model predicts threshold logits o_k :

$$\mathcal{L}_{\text{ord}} = \sum_{k=0}^{K-2} \text{BCE}(\sigma(o_k), r_k). \quad (3)$$

The stiffness head is trained with mean-squared error $\mathcal{L}_{\text{stiff}} = \|\hat{s} - s\|_2^2$. The total physical-property loss is

$$\mathcal{L}_{\text{phys}} = \lambda_m \mathcal{L}_{\text{ord}}^m + \lambda_f \mathcal{L}_{\text{ord}}^f + \lambda_s \mathcal{L}_{\text{stiff}}. \quad (4)$$

We use GradNorm to adapt $\lambda_m, \lambda_f, \lambda_s$ during training so that the three tasks contribute balanced gradients to the shared encoder.

F. Human-to-Robot Transfer

The physical-property predictor is trained primarily from human demonstrations, but it must be used on a real dexterous hand. We reduce the embodiment gap in two ways. First, the human acquisition system is designed to match the robot sensing setup: the wrist camera is aligned with the robot wrist-view camera, and the fingertip tactile sensors have the same density distribution as those on the robot hand. Second, we reduce the RGB-domain gap by translating a subset of human-hand videos into robot-hand videos using

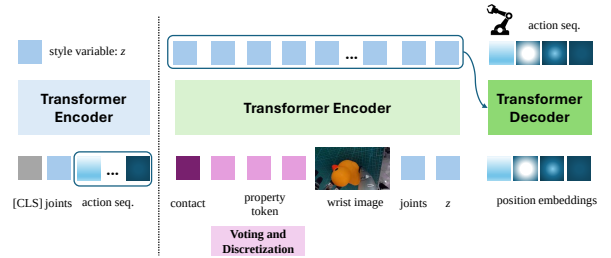


Fig. 7. Physical-property-conditioned ACT policy. Contact and predicted property tokens are discretized, voted over recent windows, and injected with wrist RGB and proprioception to generate future action chunks.

a video generation model conditioned on robot-hand visual references. An example is shown in Figure ???. This produces robot-style RGB observations while preserving the object identity and interaction dynamics.

We further co-train the predictor with real-robot trials in which the robot executes interaction motions analogous to the human demonstrations. Similar to recent human-to-robot scaling approaches [?], co-training encourages the model to learn physical-property cues that are invariant across human and robot embodiments, rather than overfitting to the appearance of the human hand.

G. Physical-Property Prediction for Policy Learning

For downstream adaptive grasping, we use the physical-property predictor as an online perception module rather than only an offline estimator. Its predictions are converted into structured inputs for an ACT-style imitation-learning policy M_{ACT} shown in Figure ??, allowing the policy to condition its future action chunk on the object’s inferred mass, stiffness, and friction. Since reliable physical-property estimates are available only after contact, we introduce a binary contact token $c \in \{0, 1\}$. Before contact, the property token is set to zero. After contact, the stiffness prediction is discretized into $s_{\text{bin}} \in \mathbb{R}^{10}$, and the physical token is formed as $\text{token}_{\text{phys}} = \text{concat}(\text{Emb}(c), \text{Emb}(m), \text{Emb}(s_{\text{bin}}), \text{Emb}(f))$.

At policy time step t , the policy observation is $o_t =$

$(I_t, p_t, \text{token}_{\text{phys},t})$, where I_t is the wrist RGB observation and $p_t \in \mathbb{R}^6$ is the dexterous-hand proprioceptive state. Following ACT [?], the policy predicts a chunk of H future actions, $A_t = (a_t, \dots, a_{t+H-1})$, rather than a single action. During training, a CVAE-style encoder maps the demonstrated action chunk and the current observation to a posterior latent distribution, $q_\phi(z_t | A_t, o_t) = \mathcal{N}(\mu_\phi(A_t, o_t), \Sigma_\phi(A_t, o_t))$. A Transformer decoder policy then reconstructs the action chunk from the observation and latent variable, $\hat{A}_t = \pi_\theta(o_t, z_t)$. The training objective is

$$\mathcal{L}_{\text{ACT}} = \frac{1}{H} \sum_{h=0}^{H-1} \|\hat{a}_{t+h} - a_{t+h}\|_1 + \beta D_{\text{KL}}(q_\phi(z_t | A_t, o_t) \| \mathcal{N}(0, I)). \quad (5)$$

The first term trains the decoder to imitate expert action chunks, while the KL term regularizes the latent action representation toward a unit Gaussian prior. At inference, the policy uses the prior latent, as in ACT, and executes action chunks with temporal ensembling. By conditioning the policy on predicted physical properties after contact, the robot can adapt grasp force, finger closure, and anti-slip behavior to objects with different mass, stiffness, and friction.

V. EXPERIMENTS

We evaluate H2Touch along two axes. First, we test whether the proposed visual-tactile predictor can estimate object mass, structural stiffness, and static friction coefficient under different generalization settings. Second, we deploy the predicted physical properties in a real-robot adaptive grasping policy and evaluate whether they improve grasp success and force adaptation.

A. Physical-Property Prediction Protocol

We use three data splits to evaluate the physical-property predictor. In the *in-distribution* split, all episodes from all objects are randomly split at the episode level into 80% training, 10% validation, and 10% testing sets. This setting measures whether the model predicts physical properties for unseen grasps of objects observed during training. In the *leave-one-class-out* split, episodes are split at the object level: all episodes of 80% of the objects are used for training, and the remaining objects are divided equally for validation and testing while ensuring that each coarse object category contributes at least one validation and one test object. This setting evaluates generalization to unseen objects from related semantic categories. Finally, in the *one-shot* split, only one object from each coarse category is used for training and all other objects are used for testing, providing a low-data stress test for category-level generalization. All reported results are averaged over three random repetitions and are computed over all temporal windows from the corresponding test episodes.

B. Physical-Property Prediction Metrics

For mass and friction classification, we report accuracy, $\text{Acc} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{y}_i = y_i]$, and Macro-F1, $\text{MacroF1} =$

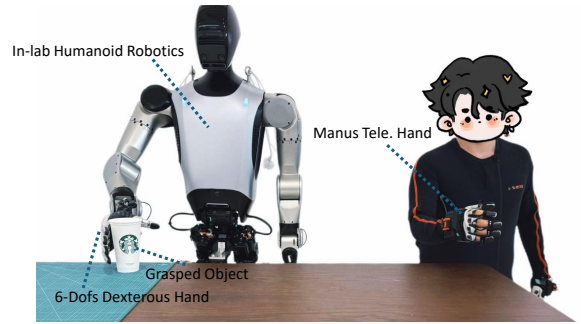


Fig. 8. Robot teleoperation setup for downstream grasping. A Manus glove controls the dexterous hand to collect demonstrations in which the robot grasps and lifts objects with property-dependent forces.

$\frac{1}{C} \sum_{c=1}^C \frac{2P_c R_c}{P_c + R_c}$. Accuracy measures the overall correct prediction ratio, while Macro-F1 averages class-wise F1 scores and therefore reflects whether the model performs well across both frequent and infrequent classes.

For stiffness regression, we report MAE, RMSE, MAPE, Pearson correlation, and Within- k . MAE and RMSE measure absolute numerical error, with $\text{MAE} = \frac{1}{N} \sum_i |\hat{s}_i - s_i|$ and $\text{RMSE} = \sqrt{\frac{1}{N} \sum_i (\hat{s}_i - s_i)^2}$. MAPE denotes mean absolute % error, $\text{MAPE} = \frac{100}{N} \sum_i \frac{|\hat{s}_i - s_i|}{|s_i|}$, and normalizes the stiffness error by the ground-truth stiffness scale. Pearson correlation measures trend consistency, $\rho = \frac{\sum_i (\hat{s}_i - \bar{\hat{s}})(s_i - \bar{s})}{\sqrt{\sum_i (\hat{s}_i - \bar{\hat{s}})^2} \sqrt{\sum_i (s_i - \bar{s})^2}}$. Within- k reports the % of predictions inside a relative-error tolerance, $\text{Within-}k = \frac{1}{N} \sum_i \mathbf{1}\left[\frac{|\hat{s}_i - s_i|}{|s_i|} \leq k\right]$. These metrics jointly measure numerical error, scale-normalized error, trend consistency, and whether the estimate is useful within an engineering tolerance.

C. Physical-Property Prediction for Policy Learning

The downstream experiments use an in-house dexterous robot platform composed of a 7-DoF manipulator and a direct-drive 6-DoF dexterous hand. The fingertips are equipped with piezoresistive tactile arrays with the same spatial density as the human wearable glove. Real-robot demonstrations are collected by teleoperation with a Quantum Manus glove shown in Figure ???. For each object, the robot starts from a fixed initial grasp pose, grasps the object, and lifts it vertically. We collect 10 expert demonstrations for each of 50 objects, giving 500 robot demonstrations in total. The demonstrations use larger forces for low-friction objects but avoid visibly crushing soft objects. This protocol focuses on physical-property adaptation rather than generalization over object placement.

We evaluate two robot object splits. The ID split contains 40 H2Touch objects that are physically suitable for the robot hand. The OOD split contains five groups of paired objects with similar RGB appearance but clearly different mass, stiffness, or static friction coefficient. This split directly tests whether the property predictor and policy can distinguish visually similar objects with different manipulation requirements.

The physical-property predictor is deployed as the perception front-end for policy learning. Once contact is detected,

recent visual-tactile observations are passed through M_{phys} , and the resulting mass, stiffness-bin, and friction predictions are used as property tokens for the ACT policy described in Sec. ?? . This setup tests whether physical-property prediction is useful as an action-conditioning signal, not only as an offline recognition task.

D. Downstream Task Metrics

A grasp trial is counted as a clean success if the robot lifts the object and holds it for more than 5 seconds without dropping it or visibly crushing it. If the object is lifted and held but a soft object is visibly compressed, we record it as an over-force success. Each method is evaluated with three repeated trials per object. For successful trials, we also compare the robot’s steady-state grasping force with human demonstrations.

For frame t , finger j , and tactile taxel set Ω_j , the normalized tactile force is $g_{t,j} = \frac{1}{255|\Omega_j|} \sum_{u \in \Omega_j} I_{\text{tac}}^{t,j}(u)$, and the steady-state force is $\bar{g} = \frac{1}{T_s J} \sum_{t \in \mathcal{T}_s} \sum_{j=1}^J g_{t,j}$, where $\mathcal{T}_s$ denotes the steady grasping frames. We report the absolute error between robot and human normalized force, the Pearson correlation over objects, and a pairwise ranking score, $\text{Pairwise} = \frac{2}{M(M-1)} \sum_{a < b} \mathbf{1}[\text{sign}(\bar{g}_a^r - \bar{g}_b^r) = \text{sign}(\bar{g}_a^h - \bar{g}_b^h)]$. Superscripts r and h denote robot and human force statistics. The pairwise score measures whether a policy preserves human-like relative force ordering across objects rather than applying an averaged grasping force.

E. Implementation Details and Baselines

The physical-property predictor is trained for 30 epochs with a batch size of 32. We use AdamW with an initial learning rate of 5×10^{-5} and a weight decay of 10^{-4} . Each training sample contains a synchronized temporal window of 15 frames. The ordinal loss uses an ordinal scale of 15. For language conditioning, we precompute BERT-base-uncased word embeddings with a maximum sequence length of 256 and use `all-mpnet-base-v2` for sentence embeddings. GradNorm is trained with a learning rate of 5×10^{-4} and $\alpha = 1.5$. For human-to-robot transfer, we co-train with human data and robot teleoperation data using the same training configuration.

For the adaptive grasping policy, continuous stiffness predictions are discretized into $K = 10$ quantile bins. We set the contact token to one when any fingertip tactile image has nonzero activation with temporal variation over a continuous 15-frame window; otherwise the contact and physical-property tokens are set to zero. Contact and property tokens are projected to a 512-dimensional policy embedding. During training, we inject Gaussian noise into the property tokens to simulate prediction fluctuations during real deployment. The ACT backbone uses four Transformer encoder layers and four Transformer decoder layers.

Real-time deployment. At deployment time, M_{phys} and the adaptive grasping policy run online. Contact is detected from the temporal variation of each fingertip tactile image over a 15-frame queue. After contact, a lightweight VLM

infers an object category name from the wrist RGB observation. We match this name to the predefined H2Touch category set and randomly select one text feature F_{text} from the matched category as the semantic prior. The physical-property predictor then receives the queued RGB, tactile, visual-flow, tactile-flow, and text features. Because single-window property predictions can fluctuate, we use a voting mechanism analogous to action chunking: for each property, the currently most probable prediction over recent windows is passed to the downstream policy.

For physical-property prediction, we ablate the tactile branch, visual branch, visual/tactile flow inputs, textual prior, and loss-balancing strategy. For downstream grasping, we compare against ACT [?], which is the backbone of our policy without explicit property conditioning, and ViTacFormer [?], an ACT-based visuo-tactile manipulation model that directly fuses visual and tactile observations.

F. Physical-Property Prediction Results

Table ?? summarizes the main physical-property prediction results. The in-distribution split gives the strongest performance, with mass and friction accuracies above 97% and stiffness MAPE below 6%. This shows that the visual-tactile-text model can reliably infer all three physical properties for unseen grasps of objects observed during training.

The leave-one-class-out split is more demanding because the test objects are never seen during training. Even under this setting, mass accuracy remains 87.5%, friction accuracy reaches 98.5%, stiffness MAPE stays below 10%, and the stiffness Pearson correlation is 0.947. These results indicate that the model learns category-level and interaction-level physical cues rather than memorizing object identities. In the one-shot split, performance decreases substantially, as expected, but the model still obtains nontrivial mass and friction classification accuracy and a stiffness MAPE of 16.84%. This suggests that a small number of category exemplars can still provide useful priors for unseen objects.

The human-to-robot transfer setting evaluates whether the predictor remains usable after moving from human demonstrations to robot sensing and interaction. With robot-style RGB translation and co-training on a small amount of robot teleoperation data, the leave-one-class-out transfer result remains close to the human-only counterpart, with only moderate degradation across the three properties. This supports our design choice of combining the large human dataset with limited robot data for downstream deployment.

G. Adaptive Grasping Results

Table ?? reports real-robot grasping outcomes. ViTacFormer has low success rates in our dexterous-hand setup and fails to lift many objects. Compared with ACT, our physical-property-conditioned policy achieves clean success that is 15.8% higher on ID objects and 49.3% higher on OOD objects. The ID-to-OOD drop is only 8.0% for our method, compared with 41.5% for ACT. This gap is most visible on the OOD split, where visually similar objects can require different grasping forces because of hidden differences in

TABLE II
PHYSICAL-PROPERTY PREDICTION RESULTS.

Setting	Mass			Stiffness					Friction	
	Acc. $\uparrow$	Macro-F1 $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	MAPE $\downarrow$	Pearson $\uparrow$	Within 10% $\uparrow$	Within 20% $\uparrow$	Acc. $\uparrow$	Macro-F1 $\uparrow$
In distribution	0.972	0.970	0.248	0.386	5.51	0.980	84.22	96.56	0.988	0.984
Leave-one-class-out avg.	0.875	0.866	0.435	0.577	9.08	0.947	61.56	89.64	0.985	0.981
One-shot avg.	0.492	0.480	0.837	1.457	16.84	0.680	49.47	75.19	0.569	0.527
Human-to-robot transfer (leave-one-class-out)	0.812	0.805	0.485	0.645	11.25	0.915	58.73	81.95	0.965	0.960

TABLE III
REAL-ROBOT GRASPING SUCCESS RATES.

Split	Method	Clean Success $\uparrow$	Over-force Success $\downarrow$
ID	ACT [?]	80.00%	6.20%
	ViTacFormer [?]	31.80%	1.50%
	Ours	95.80%	2.80%
OOD	ACT [?]	38.50%	25.60%
	ViTacFormer [?]	23.10%	2.60%
	Ours	87.80%	2.40%

TABLE IV
NORMALIZED GRASPING FORCE COMPARISON.

Comparison	Split	Method	Shared-object ratio	Abs. Err. $\downarrow$	Pearson $\uparrow$	Pairwise $\uparrow$
ACT vs. Ours	ID	ACT [?]	91.30%	0.040	0.889	0.852
		Ours		0.035	0.862	0.843
	OOD	ACT [?]	69.20%	0.076	0.438	0.639
		Ours		0.061	0.852	0.806
ViTacFormer vs. Ours	ID	ViTacFormer [?]	43.50%	0.077	0.503	0.733
		Ours		0.044	0.851	0.844
	OOD	ViTacFormer [?]	30.80%	0.116	0.637	0.667
		Ours		0.065	0.915	0.833

TABLE V
INPUT-MODALITY ABLATION.

Experiment	Mass			Stiffness					Friction	
	Acc. $\uparrow$	F1 $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	MAPE $\downarrow$	ρ $\uparrow$	W10 $\uparrow$	W20 $\uparrow$	Acc. $\uparrow$	F1 $\uparrow$
w/o tactile + text	0.660	0.648	0.756	1.152	16.70	0.769	46.39	80.69	0.915	0.905
w/o visual + text	0.660	0.640	0.668	0.853	13.12	0.887	45.97	79.79	0.844	0.862
w/o flow + text	0.661	0.662	0.858	1.366	18.91	0.633	56.94	73.47	0.898	0.857
w/o text	0.810	0.827	0.630	0.796	12.43	0.900	56.46	77.36	0.981	0.978
Ours	0.875	0.866	0.435	0.577	9.08	0.947	61.56	89.64	0.985	0.981

mass, stiffness, or friction. ACT also produces a higher over-force success rate on OOD objects, indicating that a policy without explicit property conditioning more often succeeds by applying excessive force.

Table ?? further compares successful trials by their normalized grasping force. Because failed grasps do not provide meaningful steady-state contact-force statistics, force metrics are computed only on objects that are successfully grasped by all methods in each comparison group. On ID objects, ACT and our method both produce forces reasonably close to human demonstrations. On OOD objects, however, ACT's Pearson and pairwise scores drop sharply, showing that it does not preserve human-like force ordering for unseen objects with different physical properties. Our method maintains high correlation and pairwise consistency on OOD objects, which indicates that the predicted physical properties affect grasp success through fine-grained force adaptation rather than only through coarse action selection.

H. Ablation Studies

Input modalities. Table ?? ablates the main input branches on the leave-one-class-out setting. Removing tactile input or flow input causes the largest degradation in stiffness estimation, increasing MAPE by 7.62% to 9.83% relative to the full model. This confirms that deformation, pressure evolution,

TABLE VI
LOSS AND WEIGHTING ABLATION.

Experiment	Mass			Stiffness					Friction	
	Acc. $\uparrow$	F1 $\uparrow$	MAE $\downarrow$	RMSE $\downarrow$	MAPE $\downarrow$	ρ $\uparrow$	W10 $\uparrow$	W20 $\uparrow$	Acc. $\uparrow$	F1 $\uparrow$
Manual loss weights	0.778	0.799	0.540	0.714	11.01	0.922	59.51	83.06	0.935	0.938
DWA loss weights	0.762	0.772	0.576	0.764	12.38	0.880	54.72	84.44	0.947	0.950
Cross entropy	0.635	0.657	0.534	0.730	10.07	0.945	59.38	85.14	0.979	0.978
Ours	0.875	0.866	0.435	0.577	9.08	0.947	61.56	89.64	0.985	0.981

and motion cues are central to structural stiffness estimation. Removing visual input mainly hurts friction classification, reducing friction accuracy by 13.1%, which suggests that appearance and material cues complement tactile normal-pressure sensing when tangential force is unavailable. Mass prediction requires the combination of vision, touch, and temporal flow: removing any of these branches substantially lowers mass accuracy. Without text, the visual-tactile-flow model still reaches 81.0% mass accuracy, but adding the VLM-derived semantic prior makes mass accuracy 6.5% higher and stiffness MAPE 3.35% lower. Friction prediction is already strong without text, so the gain from text is smaller.

Loss weighting and ordinal loss. Table ?? compares different multi-task training strategies. Manual weighting assigns equal weights to the three tasks, while Dynamic Weight Averaging (DWA) adjusts task weights according to the relative change of task losses across epochs. DWA does not improve over manual weighting in our setting, likely because the three physical-property tasks have different noise levels and optimization dynamics. GradNorm provides more stable balancing by equalizing gradient magnitudes on the shared encoder, making mass accuracy about 9% higher, stiffness MAPE about 2% lower, and friction accuracy about 5% higher than manual weighting.

We also compare the ordinal classification loss with standard cross entropy for mass prediction. Cross entropy treats all incorrect classes equally and ignores the ordered low/medium/high structure of physical-property ranges. Replacing ordinal regression with cross entropy severely reduces mass accuracy from 87.5% to 63.5%. Since mass is difficult to infer from visual and normal-tactile observations alone, preserving the ordinal relation between neighboring mass ranges provides important supervision.

VI. CONCLUSION

We presented H2Touch, a human visual-tactile manipulation dataset for learning object mass, structural stiffness, and static friction coefficient from real grasping interactions. Built on this dataset, our temporal multimodal predictor combines visual, tactile, motion, and language-prior cues, and can be transferred from human demonstrations to a real dexterous robot. We further showed that predicted physical

properties can serve as structured policy inputs for adaptive grasping, improving success and force adaptation on both ID and visually similar OOD objects.

Our results suggest that physical-property prediction should be treated not only as an offline perception problem, but also as a manipulation-relevant interface between contact sensing and policy learning. Future work will extend H2Touch to broader manipulation skills, richer tactile sensing, and more diverse object interactions beyond fixed-pose grasping.

REFERENCES

- [1] P. Intelligence, K. Black, N. Brown, J. Darpinian, K. Dhabalia, D. Driess, A. Esmail, M. Equi, C. Finn, N. Fusai *et al.*, “ π {0.5} : a vision-language-action model with open-world generalization,” *arXiv preprint arXiv:2504.16054*, 2025.
- [2] S. Ye, Y. Ge, K. Zheng, S. Gao, S. Yu, G. Kurian, S. Indupuru, Y. L. Tan, C. Zhu, J. Xiang *et al.*, “World action models are zero-shot policies,” *arXiv preprint arXiv:2602.15922*, 2026.
- [3] L. Kopnarski, L. Lippert, J. Rudisch, and C. Voelcker-Rehage, “Predicting object properties based on movement kinematics,” *Brain Informatics*, vol. 10, no. 1, p. 29, 2023.
- [4] T. Standley, O. Sener, D. Chen, and S. Savarese, “image2mass: Estimating the mass of an object from its image,” in *Proceedings of the 1st Annual Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 78. PMLR, 2017, pp. 324–333.
- [5] R. P. Cardoso and P. Moreno, “Estimating object physical properties from rgb-d vision and depth robot sensors using deep learning,” in *Iberian Conference on Pattern Recognition and Image Analysis*. Springer, 2025, pp. 97–110.
- [6] X. Zhang, B. L. Bhatnagar, S. Starke, I. Petrov, V. Guzov, H. Dharmo, E. Pérez-Pellitero, and G. Pons-Moll, “Force: Dataset and method for intuitive physics guided human-object interaction,” *CoRR*, 2024.
- [7] C. Wang, S. Wang, B. Romero, F. Veiga, and E. Adelson, “Swingbot: Learning physical features from in-hand tactile exploration for dynamic swing-up manipulation,” in *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2020, pp. 5633–5640.
- [8] B. Sundaralingam and T. Hermans, “In-hand object-dynamics inference using tactile fingertips,” *IEEE Transactions on Robotics*, vol. 37, no. 4, pp. 1115–1126, 2021.
- [9] Z. Xu, J. Wu, A. Zeng, J. B. Tenenbaum, and S. Song, “Densephysnet: Learning dense physical object representations via multi-step dynamic interactions,” *arXiv preprint arXiv:1906.03853*, 2019.
- [10] J. Wu, J. J. Lim, H. Zhang, J. B. Tenenbaum, and W. T. Freeman, “Physics 101: Learning physical object properties from unlabeled videos,” in *BMVC*, vol. 2, no. 6, 2016, p. 7.
- [11] Z. Guo, H. Chen, X. Mai, Q. Qiu, G. Ma, Z. Kappassov, Q. Li, and N. Chen, “Robotic perception with a large tactile-vision-language model for physical property inference,” *arXiv preprint arXiv:2506.19303*, 2025.
- [12] A. Kruzliak, J. Hartvich, S. P. Patni, L. Rustler, J. K. Behrens, F. J. Abu-Dakka, K. Mikolajczyk, V. Kyrki, and M. Hoffmann, “Interactive learning of physical object properties through robot manipulation and database of object measurements,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2024, pp. 7596–7603.
- [13] T. Sugaiwa, G. Fujii, H. Iwata, and S. Sugano, “A methodology for setting grasping force for picking up an object with unknown weight, friction, and stiffness,” in *2010 10th IEEE-RAS International Conference on Humanoid Robots*. IEEE, 2010, pp. 288–293.
- [14] A. Dutta, E. Burdet, and M. Kaboli, “Predictive visuo-tactile interactive perception framework for object properties inference,” *IEEE Transactions on Robotics*, vol. 41, pp. 1386–1403, 2025.
- [15] M. Huang, Z. Chen, W. Xu, Z. Li, Y. Zhou, G. Zhou, W. Chen, and X. Li, “Two-stream network-driven vision-based tactile sensor for object feature extraction and fusion perception,” *arXiv preprint arXiv:2510.12528*, 2025.
- [16] M. Purri and K. Dana, “Teaching cameras to feel: Estimating tactile physical properties of surfaces from images,” in *European Conference on Computer Vision*. Springer, 2020, pp. 1–20.
- [17] W. Yuan, S. Wang, S. Dong, and E. Adelson, “Connecting look and feel: Associating the visual and tactile properties of physical materials,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 5580–5588.
- [18] S. Sundaram, P. Kellnhofer, Y. Li, J.-Y. Zhu, A. Torralba, and W. Matusik, “Learning the signatures of the human grasp using a scalable tactile glove,” *Nature*, vol. 569, no. 7758, pp. 698–702, 2019.
- [19] A. Dutta, E. Burdet, and M. Kaboli, “Push to know!-visuo-tactile based active object parameter inference with dual differentiable filtering,” in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2023, pp. 3137–3144.
- [20] M. Bauza, F. Alet, Y.-C. Lin, T. Lozano-Pérez, L. P. Kaelbling, P. Isola, and A. Rodriguez, “Omnipush: accurate, diverse, real-world dataset of pushing dynamics with rgb-d video,” in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2019, pp. 4265–4272.
- [21] P. Uttayopas, X. Cheng, J. Eden, and E. Burdet, “Object recognition using mechanical impact, viscoelasticity, and surface friction during interaction,” *IEEE Transactions on Haptics*, vol. 16, no. 2, pp. 251–260, 2023.
- [22] P. Agrawal, A. V. Nair, P. Abbeel, J. Malik, and S. Levine, “Learning to poke by poking: Experiential learning of intuitive physics,” *Advances in neural information processing systems*, vol. 29, 2016.
- [23] R. Zheng, D. Niu, Y. Xie, J. Wang, M. Xu, Y. Jiang, F. Castaneda, F. Hu, Y. L. Tan, L. Fu, T. Darrell, F. Huang, Y. Zhu, D. Xu, and L. Fan, “Egoscale: Scaling dexterous manipulation with diverse egocentric human data,” *arXiv preprint arXiv:2602.16710*, 2026.
- [24] S. Kareer, D. Patel, R. Punamiya, P. Mathur, S. Cheng, C. Wang, J. Hoffman, and D. Xu, “Egomimic: Scaling imitation learning via egocentric video,” *arXiv preprint arXiv:2410.24221*, 2024.
- [25] S. Yu, K. Lin, A. Xiao, J. Duan, and H. Soh, “Octopi: Object property reasoning with large tactile-language models,” *arXiv preprint arXiv:2405.02794*, 2024.
- [26] A. Silva, M. Brites, T. Paulino, and P. Moreno, “Estimation of lightweight object’s mass by a humanoid robot during a precision grip with soft tactile sensors,” in *2019 Third IEEE International Conference on Robotic Computing (IRC)*. IEEE, 2019, pp. 344–348.
- [27] M. Bednarek, P. Kicki, J. Bednarek, and K. Walas, “Gaining a sense of touch object stiffness estimation using a soft gripper and neural networks,” *Electronics*, vol. 10, no. 1, p. 96, 2021.
- [28] M. Burgess, J. Zhao, and L. Willemet, “Learning object compliance via young’s modulus from single grasps using camera-based tactile sensors,” in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 18 535–18 542.
- [29] Y. Shuai, R. Yu, Y. Chen, Z. Jiang, X. Song, N. Wang, J. Zheng, J. Ma, M. Yang, Z. Wang *et al.*, “Pugs: Zero-shot physical understanding with gaussian splatting,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 4478–4485.
- [30] D. Guo, Y. Xiang, S. Zhao, X. Zhu, M. Tomizuka, M. Ding, and W. Zhan, “Phygrasp: generalizing robotic grasping with physics-informed large multimodal models,” in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2025, pp. 14 915–14 922.
- [31] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International conference on machine learning*. PmLR, 2021, pp. 8748–8763.
- [32] P. Y. Chen, C. Liu, P. Ma, J. Eastman, D. Rus, D. Randle, Y. Ivanov, and H. Matusik, “Learning object properties using robot proprioception via differentiable robot-object interaction,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2025, pp. 5997–6004.
- [33] Y. Li, C. Guo, J. Ren, B. Chen, C. Cheng, H. Zhang, and H. Lu, “Softgrasp: Adaptive grasping for dexterous hand based on multimodal imitation learning,” *Biomimetic Intelligence and Robotics*, vol. 5, no. 2, p. 100217, 2025.
- [34] Y. Chen, W. Xu, Z. Yu, T. Tang, Y. Li, S. Yao, and C. Lu, “Fbi: Learning dexterous in-hand manipulation with dynamic visuotactile shortcut policy,” *arXiv preprint arXiv:2508.14441*, 2025.
- [35] Z. Cheng, Y. Zhang, W. Zhang, H. Li, K. Wang, L. Song, and H. Zhang, “Omnivtla: Vision-tactile-language-action model with semantic-aligned tactile sensing,” *arXiv preprint arXiv:2508.08706*, 2025.
- [36] J. Yu, H. Liu, Q. Yu, J. Ren, C. Hao, H. Ding, G. Huang, G. Huang, Y. Song, P. Cai *et al.*, “Forcevla: Enhancing vla models with a force-aware moe for contact-rich manipulation (2025),” *arXiv preprint arXiv:2505.22159*, 2025.
- [37] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, “Learning fine-grained bimanual manipulation with low-cost hardware,” *arXiv preprint arXiv:2304.13705*, 2023.
- [38] R. Watanabe, M. Alvarez, P. Ferreira, P. Savkin, and G. Sano, “Ftact: Force torque aware action chunking transformer for pick-and-reorient bottle task,” *arXiv preprint arXiv:2509.23112*, 2025.
- [39] L. Heng, H. Geng, K. Zhang, P. Abbeel, and J. Malik, “Vitaformer: Learning cross-modal representation for visuo-tactile dexterous manipulation,” *arXiv preprint arXiv:2506.15953*, 2025.